

CPT



Cisco Packet
Tracer

Virtual LAN

Network Protocols and Security



Cisco Packet Tracer

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Software Download

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Download the software (Mac/Windows/Ubuntu):

<https://www.netacad.com/resources/lab-downloads>

- Launch the Packet Tracer install program.
- When prompted, click on Skills For All green button to authenticate.
- More details: https://www.netacad.com/skillsforall/files/Cisco_Packet_Tracer_Download_and_Installation_Instructions.pdf
- Starter Course: <https://www.netacad.com/courses/getting-started-cisco-packet-tracer>
- For Built-In Examples, Go to: File → Open Samples

Physical Cabling

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- Click Connections (lightning bolt icon) to see types of cables
- Unlike devices are connected using *Copper straight-Through* cable
 - PC to Wireless Router
 - Wireless Router to cable modem
- Like devices are connected using *Copper Cross-Over* cable
 - Switch to Switch
- After the devices negotiate connection-Both ends are green
- Click >>> (left-bottom) for faster negotiation (orange turns to green)
- If the connection is red- Go in and activate the interface

Connecting Laptop to WiFi

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Replace the wired Ethernet network interface card (NIC) with a wireless NIC and connect the Laptop to the router wirelessly.

- Click Laptop, and select the Physical tab
- In the Physical tab, you will need to remove the Ethernet copper module and replace it with the Wireless WPC300N module.
 - Power off Laptop by clicking the power button on the side of the laptop
 - Remove the currently installed Ethernet copper module by clicking on the module on the side of the laptop and dragging it to the MODULES pane on the left of the laptop window
 - Install the wireless WPC300N module by clicking it in the MODULES pane and dragging it to the empty module port on the side of the Laptop.
 - Power on the Laptop by clicking the Laptop power button again.

Connecting Laptop to WiFi

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- With the wireless module installed, connect the Laptop to the wireless network. Click the Desktop tab and select the PC Wireless
- Select the Connect tab. After a slight delay, the wireless network **HomeNetwork** will be visible in the list of wireless networks.
- Click Refresh if necessary to see the list of available networks. Select the **HomeNetwork**. Click Connect.

IP Address

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Go through the article:

[https://learn.microsoft.com/en-us/troubleshoot/
windows-client/networking/
tcpip-addressing-and-subnetting](https://learn.microsoft.com/en-us/troubleshoot/windows-client/networking/tcpip-addressing-and-subnetting)

Managing a router from terminal

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- Select console cable from connections
- Connect RS232 on laptop to console on the Router
- Now you can manage the Router using terminal emulation software

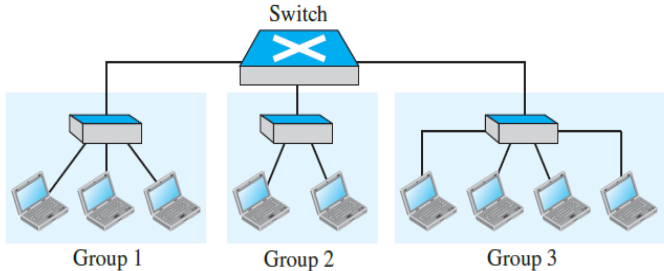
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- LAN in an engineering firm in which 10 stations are grouped into three LANs that are connected by a switch.
- if the administrators needed to move two engineers from the first group to the third group, to speed up the project being done by the third group?
- Change LAN configuration? Do rewiring?

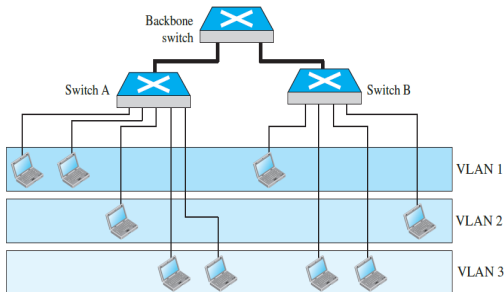
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- The idea of VLAN technology is to divide a LAN into logical segments, instead of physical segments
- If a person moves from one group to another, there is no need to change the physical configuration
- The group membership in VLANs is defined by software, not hardware

Virtual LAN

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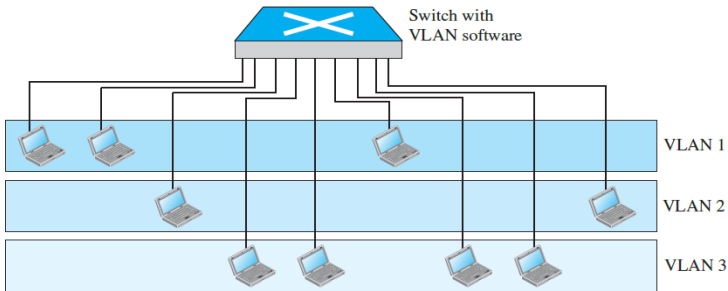


Figure: Switch using VLAN software